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Early Carboniferous paleomagnetic results from the northeastern margin of the Qinghai–Tibetan plateau and their implications

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ABSTRACT

We conducted paleomagnetic investigations on limestone from the Lower Carboniferous Huaitoutala Formation in the Qaidam Basin near Delingha City, Qinghai Province, China. The characteristic remanent magnetization ($D = 5.8°$, $I = -25.7°$, $k = 114.3$, $\alpha_{95} = 4.8°$) passes a fold test and indicates a paleopole position of −39.2°N, 90.4°E and a paleolatitude of 13.5°N for the Qaidam Block for the early Carboniferous. Based on global tectonic reconstructions and paleontological evidence, we suggest that the Qaidam Block was adjacent to, but independent from, the North China, South China, Alashan–Hexi and Tarim blocks at this time. This result suggests that Pre-Carboniferous sutures reported around the Qaidam Basin represent collisional events within Gondwana, rather than the final sutures that gave rise to the present tectonic configuration.

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1. Introduction

Most Southeast Asian continental terranes originated from the Indian-Western Australian margin of eastern Gondwana. The separation and northward migration of these blocks from Gondwana occurred in three phases that were linked with the successive opening and closure of three intervening Tethyan oceans (Metcalfe, 2013). Following a long history of separation, northward drift and continent accretion, these terranes began to collide during the Late Permian and reached their final configuration within Pangea at the end of the Triassic (Metcalfe, 2006; Rakotosolofo et al., 2006; Stampfli et al., 2013). Several tectonic reconstructions for the Late Paleozoic indicate that the East Asian tectonic blocks were not attached to Pangaea and had a scattered paleogeographic distribution until their convergence in the Early Mesozoic (Scotese, 2001; Metcalfe, 2006, 2011, 2013). Paleomagnetic research that focused on the main tectonic blocks of East Asia, including the North China Block (NCB), the South China Block (SCB), and the Tarim Block, has yielded a complete and reliable Phanerozoic apparent polar wander path (Lin et al., 1985; Zhao and Coe, 1987; Yang et al., 1992, 1998; Yang and Besse, 2001; Wu et al., 1998; Fang et al., 1998; Huang et al., 2000, 2008). However, the tectonic history of smaller yet important Asian terranes remains unclear because of tectonic complexities and insufficient Paleozoic paleomagnetic data. Such data are particularly needed for the Carboniferous, which coincided with the middle part of the northward drift phase of some smaller tectonostratigraphic terranes.

The small Qaidam Basin is located on the northeastern margin of the Qinghai–Tibetan Plateau, and is surrounded by the Tarim, Alashan–Hexi, NCB and Qiangtang blocks (Fig. 1a). The Paleozoic tectonic evolution of this region and relationships between the Qaidam and the surrounding blocks are poorly understood. For example, the spatial–temporal relationship between the Qaidam and Tarim blocks is debated. Heubeck (2001) suggested that the Qaidam, Tarim and Alashan blocks collided and amalgamated before the Middle Devonian. Metcalfe (2006) suggested that the Qaidam Block was originally part of the Tarim Block because of the close similarities of their basement rocks, which consist of Early Proterozoic metamorphic rocks with a Late Proterozoic–Paleozoic sedimentary cover. However, Xu et al. (2011) reported different Late Carboniferous paleolatitudes for these two blocks. The tectonic reconstructions proposed by Cocks and Torsvik (2013) indicate that Tarim Block lay farther to the west of the NCB and SCB, and that the Qaidam Block collided with the NCB after the Devonian. Furthermore, it has been proposed that the Alashan–Hexi Block has been a western extension of the NCB since the Ordovician (Huang et al., 2001). Song et al. (2013) and Xiao et al. (2009) suggested that the Qaidam Block had already accreted to the Alashan/Hexi Corridor terrane by the Late Devonian, whereas Yuan and Yang (2015a) suggested that the Alashan Terrane, which bridged the South Central Asian Orogenic Belt, the Tarim and Qaidam blocks, and the North China

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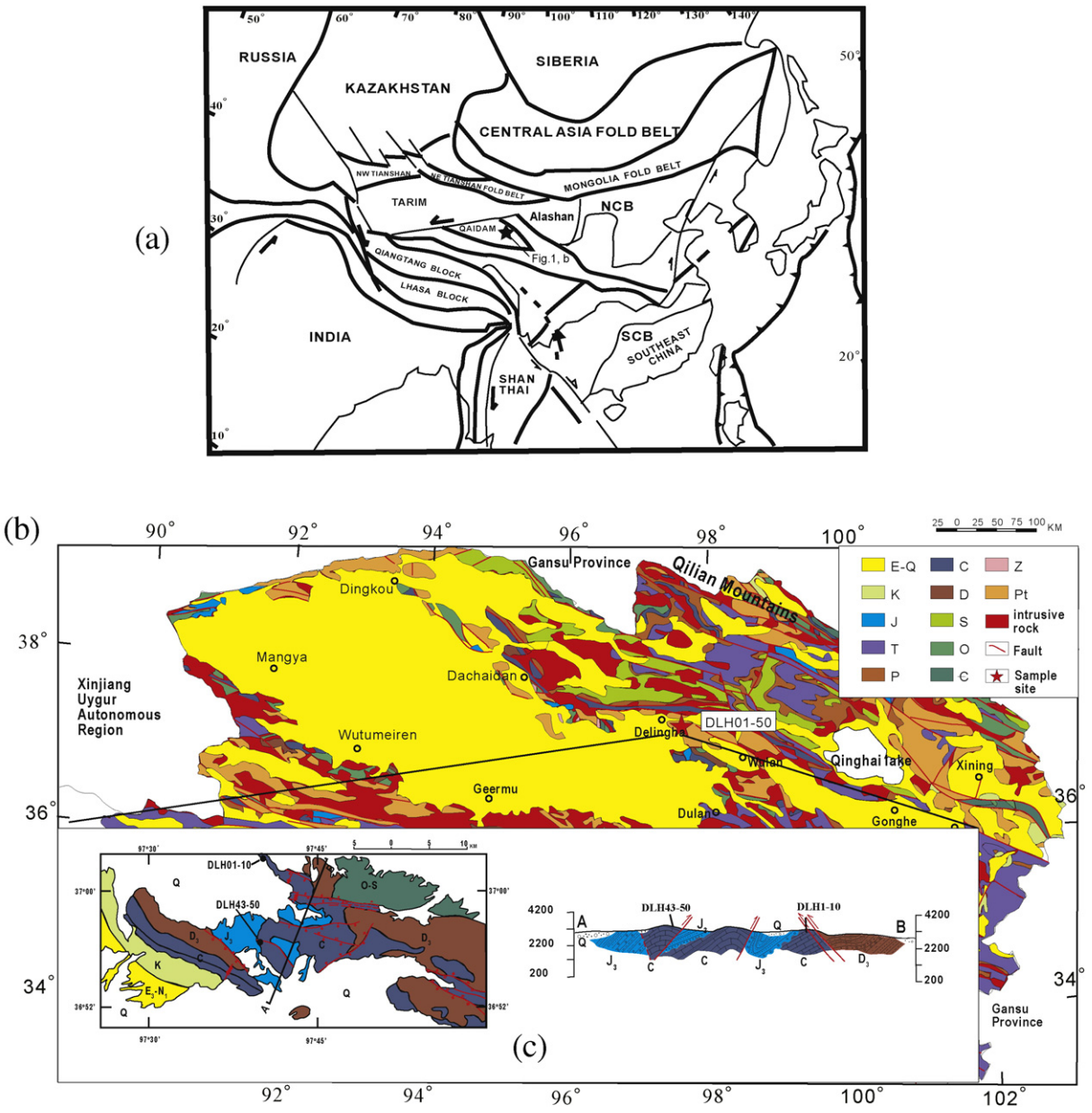


Fig. 1. (a) Simplified tectonic map of China and adjacent area (modified from Yang et al., 1992). The star indicates the sampling location. (b) Structural outline of the Qaidam Basin and its neighboring areas (modified from Ma et al., 2002). (c) Simplified geological map of the sampled location and geological cross-sections through the study area (modified from an unpublished 1:200,000 geological map).

Craton, was not part of North China by the Late Devonian. This small terrane collided with the NCB after the Early–Middle Triassic along a tectonic boundary located between the Helanshan and Zhuozishan mountains (Yuan and Yang, 2015b). Thus, many questions remain about the post-Devonian regional tectonic evolution of Asia, when the continental blocks that presently compose China were drifting northward.

The Qaidam Block is an important geological unit within this broader regional context, and its Paleozoic reconstruction should assist in understanding the evolution of surrounding tectonic belts, the break-up of Gondwana, and the convergence of Pangea. To establish the location of the Qaidam block during the Paleozoic, we carried out paleomagnetic analyses of the Carboniferous strata from the Qaidam Basin.

2. Geological setting

The Qaidam Basin is a small rhomb-shaped basin located on the northeastern margin of the Qinghai–Tibetan Plateau and is surrounded by the Altye, Qilian and Eastern Kunlun mountains. The complex tectonic history of this basin, which involved alternating periods of compressional and extensional tectonics, resulted from its paleogeographic relationship with the surrounding tectonic blocks. The basin is composed of three tectonic units: the Northern Fault Zone, the Central Fold Zone and the Southern Fault Zone. The Northern and Southern Fault zones consist of thrust faults that dip to the south, while the Central Fold Zone is covered by Mesozoic and Cenozoic strata (Chen et al., 2010). Our sampling area is located near Delingha City, and is located on the eastern side of the Central Fold Zone. This region has been subjected to five tectonic events from the Carboniferous to the Cretaceous, as follows: extension during the Carboniferous and Permian, extrusion and uplift during the Triassic, fault-bounded basin formation in the Early to Middle Jurassic, subsidence in the Late Jurassic, and basin inversion during the Cretaceous (Shang et al., 2014). In this area, Carboniferous strata have angular unconformable contacts with overlying Middle Jurassic strata and underlying Late Devonian strata.

The Carboniferous stratigraphy of the area consists of the Lower Carboniferous Amunike, Chuanshangou, Chengqianggou, and Huaitoutala formations, which are unconformably overlain by the Upper Carboniferous Keluke and Zhabusagaxiu formations (BGMRQP, 1991). The Huaitoutala Formation, which is the focus of the present study, is composed mainly of limestone and bioclastic limestone interbedded with sandstone and shale. This formation contains age diagnostic corals (*Carruthersella*, *Lithostrotionidae*, *Aulophyllidae*, and *Aulina rotiformis–Lithostrolion irregulare*) and brachiopods (*Antiquatonia insculpta–Gigantoproductus moderatiformis*, *Gigantoproductus latissimus–Kansuella kansuensis*, and *Gigantoproductus edelburgensis–Semiplanue semiplanus* groups) that are indicative of a late Visean stage (BGMRQP, 1991; Chen et al., 2003).

3. Paleomagnetic sampling and methods

Paleomagnetic samples were drilled from 18 sites in the Huaitoutala Formation, in quarries located by road no. 209 near Delingha City,

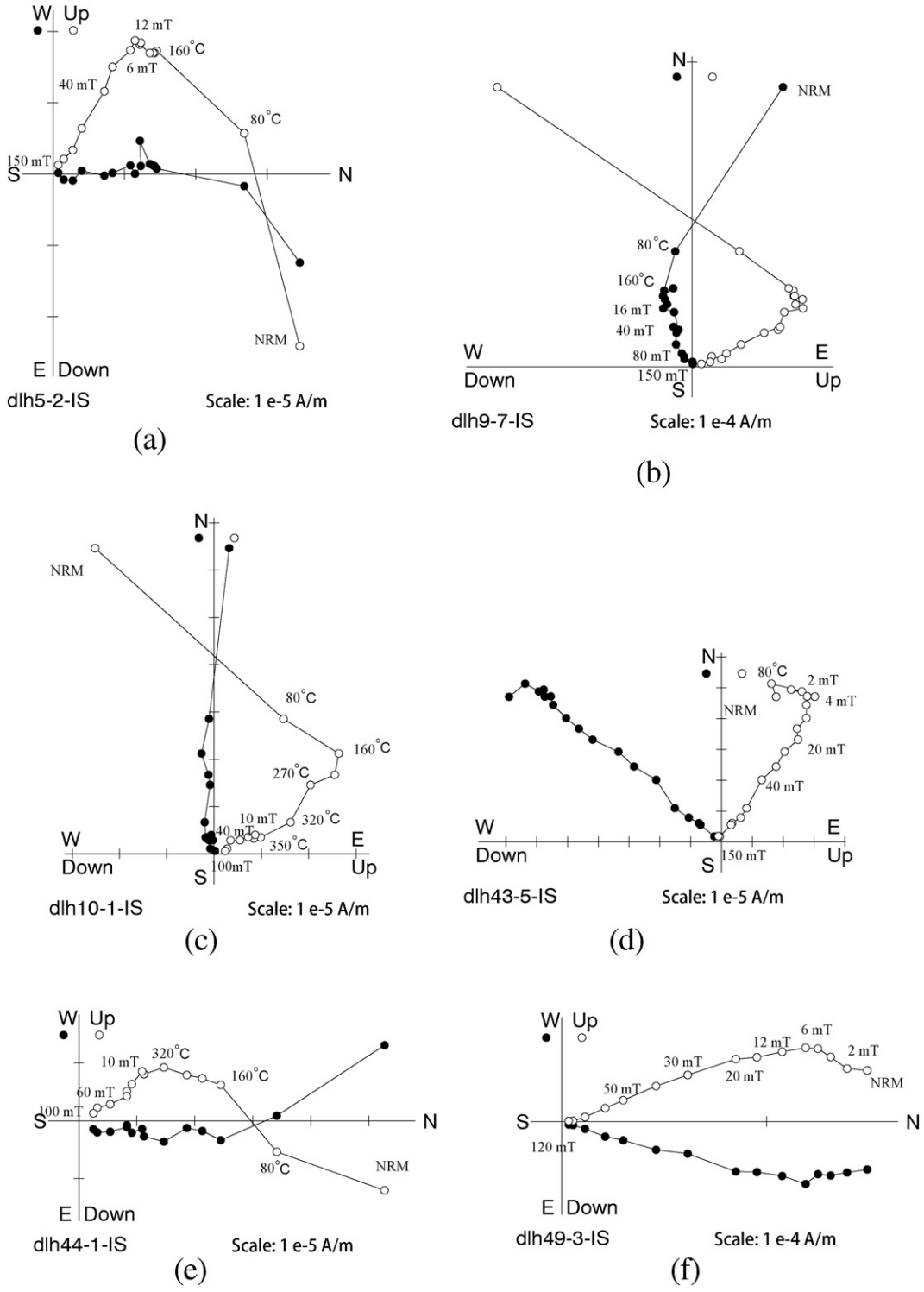


Fig. 2. Representative vector demagnetization diagrams for thermal and AF demagnetization of the natural remanent magnetization (NRM) from the studied Carboniferous samples (in situ coordinates) from the Qaidam basin.

Qinghai Province (Fig. 1). In this area, Devonian, Carboniferous, Jurassic, Cretaceous and Cenozoic strata are exposed in this region (Fig. 1c). In addition to the angular unconformities that separate Devonian, Carboniferous and Jurassic strata, the Cenozoic, Cretaceous and Jurassic strata have disconformable relationships. As mentioned above, this area underwent five major phases of tectonism, including three phases of collision and extrusion tectonics during the Triassic, Early Cretaceous and Later Cretaceous, respectively. Consequently, folds in the sampling area have undergone multiple stages of development. The sampled Carboniferous strata were deformed in the Triassic at least, but likely underwent additional deformation phases. The orientation of strata on both sides of the sampled folds is generally parallel, which suggests that folding was cylindrical, and, therefore, the application of a straightforward tilt correction to our paleomagnetic data is adequate.

The present-day geomagnetic declination at the sampling sites (−0.2° at Delingha City) was estimated using the International Geomagnetic Reference Field (IAGA, 2010). All drilled samples have a diameter of 2.5 cm and were cut into 2.2-cm-long specimens, and analyzed at the paleomagnetic laboratory of the South China Sea Institute of Oceanology (Chinese Academy of Sciences). Duplicate specimens were selected for acquisition of isothermal remanent magnetization (IRM) and for thermal demagnetization of composite IRMs (Lowrie, 1990). These analyses were performed at the paleomagnetism laboratory of the Australian National University in Canberra, Australia, to identify the magnetic minerals responsible for the measured paleomagnetic signals. Different fields were applied along three orthogonal sample axes (0.12 T, 0.4 T and 2.2 T along the X, Y and Z sample axes, respectively) following Lowrie (1990).

Stepwise alternating field (AF) demagnetization was carried out for some samples, while others were thermally demagnetized to 350 °C to remove high coercivity secondary minerals, such as goethite, which was followed by AF demagnetization to 80–120 mT. The natural remanent magnetization (NRM) was measured using a 2G Enterprises model 755 cryogenic magnetometer installed within a magnetically shielded laboratory. Characteristic remanent magnetization (ChRM) directions were determined using principal component analysis (Kirschvink, 1980), and site-mean directions were calculated using Fisherian statistics (Fisher, 1953). Paleomagnetic data were analyzed using the software packages developed by Enkin (2003) and Cogne (2003).

4. Results

Among the 213 samples collected from the Huaitoutala Formation, 56 contain two stable magnetization components after AF demagnetization (Fig. 2). The rock magnetic results indicate that magnetite is the dominant magnetic mineral (Fig. 3). The low-temperature/low-coercivity component (A component) was isolated at around 160–320 °C or 6–9 mT (Fig. 2). The site mean directions for this component are as follows: $D_g/I_g = 3.1°/46.7°$, $k = 54.1$, $\alpha_{95} = 5.7°$, $D_s/I_s = 350.0°/61.3°$, $k = 8.6$, $\alpha_{95} = 15.0°$ (Fig. 4a, b). These directions do not pass the fold test (McFadden, 1990; Enkin, 2003) and are close to the present-day field direction (D/I = −0.2°/56.1°). Thus, this component is interpreted to be a viscous remanent magnetization acquired in the present-day field. The second component (B component) was isolated up to a maximum applied field of 120–150 mT, and its site mean direction is $D_g/I_g = 6.7°/-33.7°$, $k = 14.7$, $\alpha_{95} = 13.9°$; $D_s/I_s = 5.8°/-25.7°$, $k = 114.3$, and $\alpha_{95} = 4.8°$ (Fig. 4c, d; Table 1). The B component directions pass the fold test (McFadden, 1990; Enkin, 2003). The parameters for the McFadden (1990) fold test are $\xi_{1(\text{in situ})} = 5.325$ in geographic coordinates and $\xi_{1(\text{tilt corrected})} = 1.638$ after tilt correction, while the critical value is $\xi_c = 3.497$ at 95% and $\xi_c = 4.849$ at 97%. In the fold test of Enkin (2003), optimal clustering of paleomagnetic directions for the Carboniferous data occurs at an unfolding of 105 ± 40%. This mean direction is distinct from younger results reported for the Qaidam Block (Neogene: 26.2°/40.0°, $\alpha_{95} = 10.5°$; Eocene: 0.6°/50.9°, $\alpha_{95} = 4.5°$; Early Cretaceous: 40.4°/34.8°, $\alpha_{95} = 7.5°$; and Late Jurassic: 20.7°/36.7°, $\alpha_{95} = 6.7°$; Chen et al., 2002; Late Permian: 333.9°/41.7°, $\alpha_{95} = 6.2°$; Xu et al., 2011). On the basis of the positive fold test result, the B component identified in the studied samples is interpreted to represent a primary magnetization component.

5. Discussion

Our sample sites were located in the lower–middle part of the Huaitoutala Formation, which corresponds to the middle–upper part of the Carboniferous Visean stage (BGMRQ, 1991; Chen et al., 2003). The available paleontological evidence from the studied limestone, does not allow a more precise age estimate. Our paleomagnetic results produce equatorial paleolatitudes of ±13.5° without any further constraints on the polarity of these paleomagnetic directions. Therefore, we propose two potential interpretations for our results. The first assumes that the paleomagnetic directions from the Huaitoutala Formation have normal polarity, and indicates that the Qaidam Block was located in the southern hemisphere during the Carboniferous, with the data corresponding to a paleopole position of 39.2°N, 270.4°E, and a paleolatitude of 13.5°S. The other possibility is that the paleomagnetic directions have reversed polarity, in which case the Qaidam Block would have been located in the northern hemisphere in the Carboniferous. The paleopole position would, thus, have been 39.2°N, 90.4°E, with a paleolatitude of 13.5°N. During the middle–late Visean stage, the

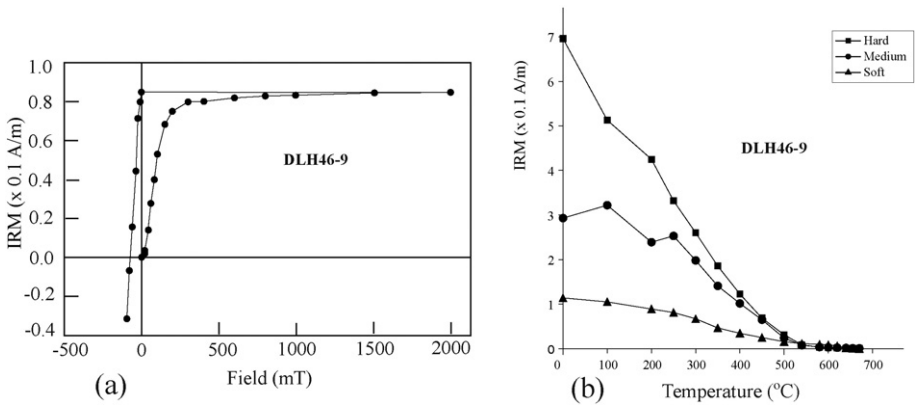


Fig. 3. (a) Isothermal remanent magnetization (IRM) acquisition curves and (b) thermal demagnetization of a three-axis IRM. (a) The IRM acquisition curve increases rapidly below 150 mT, but does not reach saturation even at 1000 mT. (b) The thermal demagnetization of a three-axis IRM indicates major unblocking at around 580 °C, which suggests that magnetite is the dominant magnetic mineral in this sample, with a smaller magnetic contribution from high-coercivity phases. Unblocking of the high coercivity component up to 100 °C indicates the likely presence of goethite, while gradual unblocking throughout the full temperature range and minor unblocking between 580 °C and 680 °C indicates a possible small magnetic contribution from hematite.

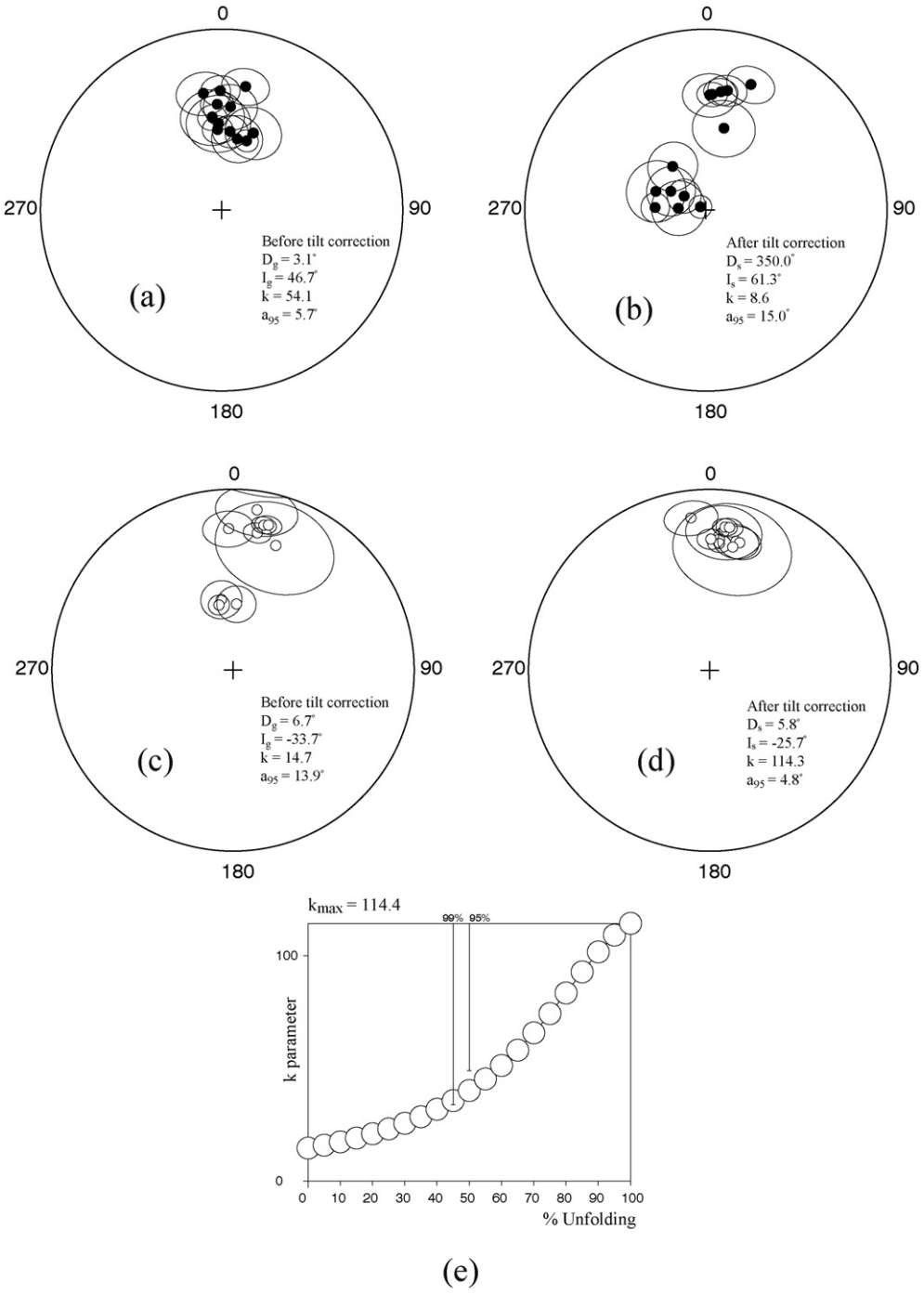


Fig. 4. Equal-area stereographic projections for site-mean directions from the studied Carboniferous samples before and after tilt correction for the (a, b) A component and the (c, d) B temperature component. (e) Progressive unfolding of the mean direction from the Carboniferous samples reveals a maximum clustering of directions at 100% unfolding.

geomagnetic field was mainly in a state of reversed polarity, with several short-term intervals of normal polarity that lasted less than 0.5 Ma (Davydov et al., 2012). Furthermore, Klootwijk (2013) suggested that the northern part of the Australian continent (i.e., the New Guinean cratonic promontory) was located in the northern hemisphere (up to paleolatitudes of 40°N around 332 Ma) between 351 Ma and 328.5 Ma, which spanned the late Tournaisian, Visean and early Serpukhovian. Also, most of the Southeast Asian continental terranes originated from the Indian-Western Australian margin of eastern Gondwana (Metcalfe, 2013).Thus, of the two options, it appears more plausible that in the early Carboniferous the Qaidam Block was located in the northern hemisphere and had a paleolatitude of 13.5°N.

We now compare our results with the coeval data from the Alashan Terrane, and the NCB, SCB and Tarim blocks (Table 2). Unfortunately, the Carboniferous paleomagnetic results reported from these blocks are limited, as are the data from the NCB and SCB (Wu et al., 1990; Zhang et al., 2001). The existing data indicate that the Tarim Block had the highest latitude of these blocks. Its paleolatitude was 18.6°N in the Early Devonian and 27.9°N in the Late Carboniferous (Fang et al., 1996). The Alashan–Hexi Block was close to the equator throughout the Carboniferous (Huang et al., 2001), and the SCB and NCB were located at low latitudes during the Early to Late Carboniferous (Zhang et al., 2001; Wu, et al., 1990). Therefore, it seems that the NCB, SCB, Alashan–Hexi and Qaidam blocks were located close to each other but

Table 1
Characteristic remanent magnetization directions for Early Carboniferous strata in the Delingha area.

Site	Latitude	Longitude	Strike/dip	N/n	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)
DLH05	37°2.305′	97°39.843′	128/36	7/13	350.6	−57.5	11.5	−28.0	47.2	8.9
DLH09			115/33	9/10	347.8	−59.8	4.4	−30.4	128.0	4.6
DLH10			116/33	8/12	3.0	−60.0	13.2	−28.4	43.4	8.5
DLH43	36°56.483′	97°39.788′	20/16	6/12	358.2	−22.8	353.0	−16.2	48.6	9.7
DLH44	36°56.589′	97°39.670’	2/15	3/12	18.9	−28.0	10.6	−31.3	28.5	23.5
DLH45			310/15	4/12	8.5	−11.4	5.9	−24.0	41.5	14.4
DLH47	36°56.599′	97°39.677’	353/20	4/12	10.0	−24.0	0.5	−28.3	301.8	5.3
DLH49	36°56.605′	97°39.697’	1/16	7/13	12.0	−19.2	6.1	−21.5	169.5	4.6
DLH50			1/16	8/12	13.7	−18.6	7.9	−21.4	123.6	5.0
Mean				9	6.7	−33.7	5.8	−25.7	114.3	4.8

Latitude and longitude for the sampling sites; N/n: number of samples used to calculate the mean/measured paleomagnetic direction; D_g, I_g, D_s, I_s: paleomagnetic declination and inclination in geographic and stratigraphic coordinates, respectively; k: precision parameter; α_{95}: half angle of the cone of 95% confidence about the mean direction after tilt correction. The site-mean direction for the higher coercivity component passes the fold test (McFadden, 1990; Enkin, 2003). Parameters for the McFadden (1990) fold test are $\xi_{1(\text{in situ})} = 5.325$ in geographic coordinates and $\xi_{1(\text{tilt corrected})} = 1.638$ after tilt correction, while the critical value is $\xi_c = 3.497$ at 95% and $\xi_c = 4.849$ at 97%. In the fold test of Enkin (2003), optimal clustering of directions for the Carboniferous data occurs for untilting of 105 ± 40%.

did not collide during the Carboniferous. However, additional data are required to accurately determine the locations of the NCB and SCB.

Lithofacies relationships suggest that in the Carboniferous the Qaidam Basin was a rifted margin basin in a stable shallow marine environment as indicated by the deposition of neritic facies carbonate rocks, and lagoonal, fan-delta (or braided river delta), transitional and coal-bearing paludal facies (Guo et al., 2002; Zhu et al., 2009; Chen et al., 2010). During the early Paleozoic, magmatic activity occurred mainly along the northeastern and southeastern margins of the Qaidam Basin, and was probably related to subduction along the northern margin of the Qaidam Block (Chen et al., 2010). Magmatism ceased in the Carboniferous and started again in the Late Permian to Early Triassic. Plagioclase granites (293.6–270 Ma) are present in the Lenghu area in the northwest of the Qaidam Basin (Gehrels et al., 2003a), and magmatic activity expanded eastward throughout the basin due to lower crustal melting associated with subduction (Chen et al., 2010). In the Late Permian, the paleolatitude of the Qaidam Block was consistent with that of the Tarim Block (Xu et al., 2011), while the NCB and SCB were close to the equator. Thus, the substantial magmatic activity in the Qaidam Block is likely to have been related to collision between the Qaidam and Tarim blocks.

The Carboniferous fossil assemblages of the Qaidam Block are similar to those of the NCB, SCB and Tarim blocks (Li and Zhang, 1999; Wang et al., 1989), although the similarity coefficient is highest with the SCB, and lowest with the Tarim Block. Based on published tectonic reconstructions (Cocks and Torsvik, 2013; Huang et al., 2001; Klootwijk, 2013; Metcalfe, 2006, 2013; Qiao and Shen, 2014; Scotese, 2001; Stampfli et al., 2013), lithofacies constraints on paleogeography and paleontological evidence, we propose a paleogeographic reconstruction for the Early Carboniferous (Fig. 5) that is modified from the reconstruction proposed by Metcalfe (2006).

Our Early Carboniferous results from the Qaidam Block allow us to discuss the tectonic relationship between this small landmass and surrounding tectonic blocks. First, the Tarim and Qaidam blocks were independent of each other during the Early Carboniferous. These two blocks might have been in close proximity since the Late Permian, which means that the initial activity on the Altyn Tagh Fault occurred no earlier than the Late Permian (Xu et al., 2011). Second, the Qaidam landmass was independent of the NCB during the Early Carboniferous. Even until the Late Permian, the paleolatitudes of the Qaidam Block and the NCB were 24°N and 14.2°N, respectively (Yang et al., 1998; Xu et al., 2011). This means that the Devonian suturing within the Qilian tectonic belt (Xiao et al., 2009), which occurred between the Qaidam Block and the NCB, does not represent the final convergence between these blocks.

As discussed by Metcalfe (2013), the separation and northward migration of the Asian blocks from Gondwana occurred in three phases that were linked with the successive opening and closure of three intervening Tethyan oceans, the Paleo-Tethys (Devonian to Triassic), the Meso-Tethys (Late Early Permian to Late Cretaceous) and the Ceno-Tethys (Late Triassic to Late Cretaceous). During these successive phases, subduction-related volcanic arcs accreted through the closure of multiple Tethyan and back-arc ocean basins, now represented by suture zones that consist of ophiolites, accretionary complexes and remnants of ocean island arcs. Furthermore, the reconstructions of eastern Gondwana for the Cambrian to Early Devonian (Metcalfe, 2006; Cocks and Torsvik, 2013) suggest that the NCB, SCB and Tarim blocks migrated along the Gondwana margin offshore of Western Australia, and that coeval island arcs, ophiolites, and accretionary complexes were preserved on the margins of these terranes (Wilde et al., 2003; Findlay, 1998; Xiao et al., 2009; Yin et al., 2007; Daukeev et al., 2002; Liu et al., 2009; De Jong et al., 2006). Arenas et al. (2014) suggested that two high pressure events were associated with the opening of an ephemeral ocean basin in Early Devonian times, during the birth of Pangaea. Similar processes probably occurred during the complex tectonic evolution covering the accretion of Gondwana and the subsequent dispersion of continental fragments. Although coeval petrological evidence could provide clues concerning Paleozoic reconstructions, the major questions concerning the interaction of East Asian tectonic blocks concern the location of the main convergent boundaries and the timing of collisions. Our initial results from Carboniferous strata constrain the relative position of the Qaidam Block with respect to the NCB, SCB and Tarim blocks. Further

Table 2
Carboniferous paleomagnetic poles from the main tectonic blocks of China.

Age	Sample area	N	Paleopole		A95 (°)	P_{lat}	Reference
			λ (°N)	Φ (°E)			
C1	Delingha, Qaidam	9	−39.2	90.4	4.4	13.5°N	This study
C1–2	Zhongwei, Alashan–Hexi	10	10.5	14.0	6.2	6.0°S	Huang et al. (2001)
C3	Zhongwei, Alashan–Hexi	5	33.3	10.2	16.7	5.4°N	Huang et al. (2001)
C3	Hancheng, North China	3 (15)	11.9	30.0	5.5	10.1°N	Wu et al. (1990)
C1	Guizhou, South China	3 (24)	47.5	229.1	9.6	0.2°N	Zhang et al. (2001)
C3	Keping, Tarim	8	46.6	170.2		27.9°N	Fang et al. (1996)

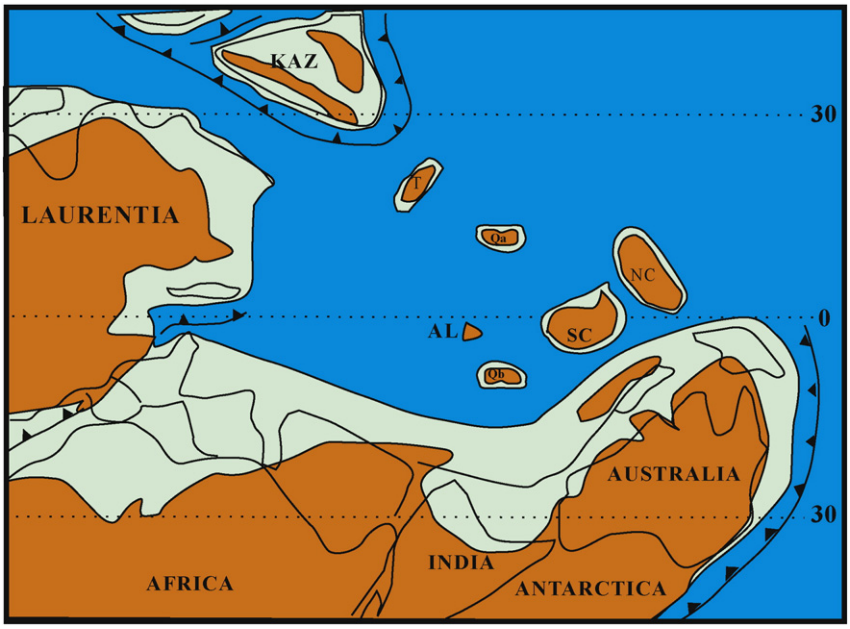


Fig. 5. Schematic global paleogeographic reconstruction of the Qaidam and nearby tectonic blocks during the Early Carboniferous (modified from Metcalfe, 2006). Relative longitudes of the blocks are unconstrained. Abbreviations are: AL = Alashan–Hexi corridor; KAZ = Kazakhstan; NC = North China Block; Qa, Qb = alternative positions of Qaidam Block as discussed in the main text; SC = South China Block; T = Tarim Block.

research on strata from other geological periods and development of a reliable Paleozoic apparent polar wander path for the Qaidam Block would help to resolve these uncertainties.

6. Conclusions

We conducted paleomagnetic research on the Carboniferous limestones from Qinghai Province, China, and obtained a characteristic remanent magnetization of D = 5.8°, I = −25.7°, and α_{95} = 4.8° that passes a fold test. A paleopole postion of −39.2°N, 90.4°E and paleolatitude of 13.5°N were obtained from our Early Carboniferous data for the Qaidam Block, assuming that the paleomagnetic direction recorded a reversed polarity interval. Considering plate reconstructions and paleontological evidence, we suggest that the Qaidam landmass was close to, but independent of, the NCB, SCB and Tarim blocks. Although the timing of suturing has been reported for the margins of the Qaidam landmass, we suggest that these pre-Carboniferous collisional events occurred when the Qaidam Block was still part of Gondwana, or that they occurred during older accretionary events associated with the northward drift of these small terranes. The final suturing that gave rise to the present configuration of continental blocks is not clearly evident in the geological record.

Acknowledgments

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